

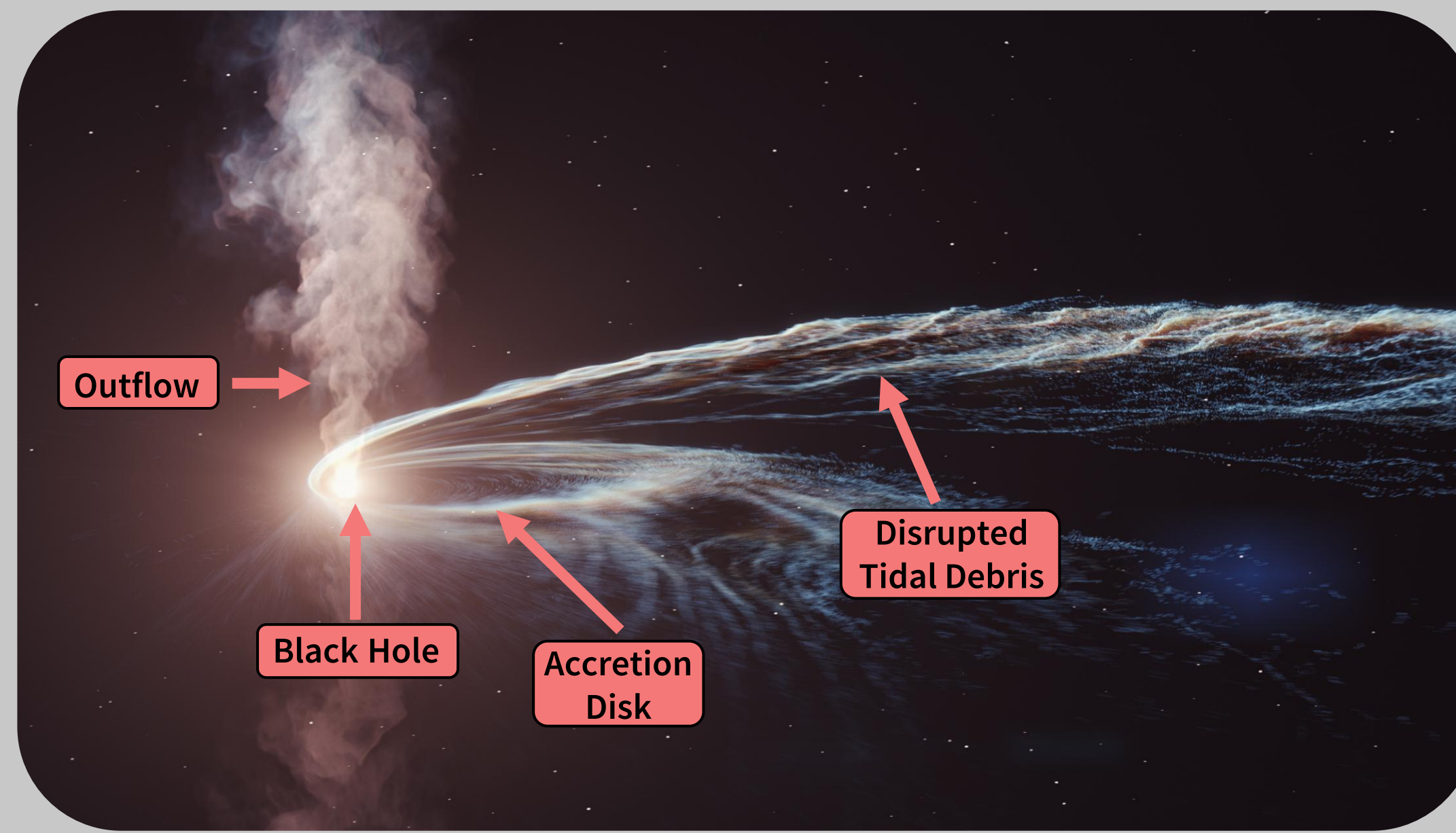
Not Every Black Hole Stays Home

Using Tidal Disruption Events (TDEs) to Search for the Wanderers

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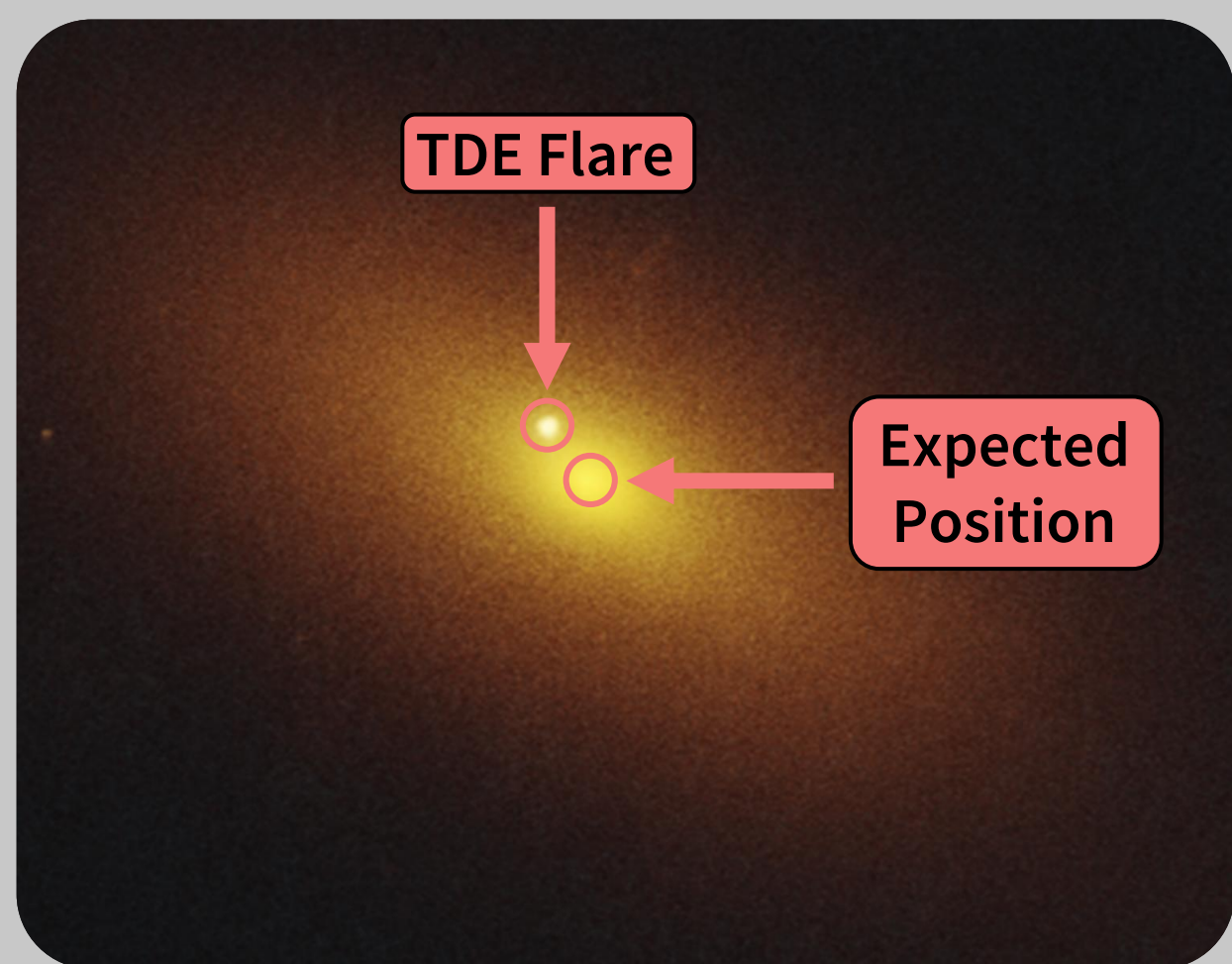
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Artist's concept of a tidal disruption event, depicting TDE 2019dsg, one of the 14 TDEs in this study's sample. Labels show the black hole, disrupted stellar debris, accretion disk, and outflow.

Introduction

- Nearly all large galaxies host a central SMBH [1]
- Mergers can leave an additional **wandering black hole** in the galaxy [2]
- Wandering black holes generally lack an accretion disk, hard to detect [2]
- Tidal disruption events (TDEs)** occur when a star gets ripped apart by a black hole's gravity, briefly revealing it [3]
- TDEs are assumed nuclear; some are **off-nuclear (offset)** [3, 4, 5, 6]
- Offset TDEs are **measurably separated from the host center**
- Offset TDEs are a **rare direct probe** of wandering black holes [2]



Hubble Space Telescope image of TDE 2024tvd, a tidal disruption event offset 0.914" (2,600 light-years) from its host galaxy's nucleus. [4] Taken with WFC3/UVIS in the F275W and F625W filters. The first off-nuclear optically-selected TDE, and part of the sample studied in this work. The TDE appears as the white point source; the host galaxy is shown in orange.

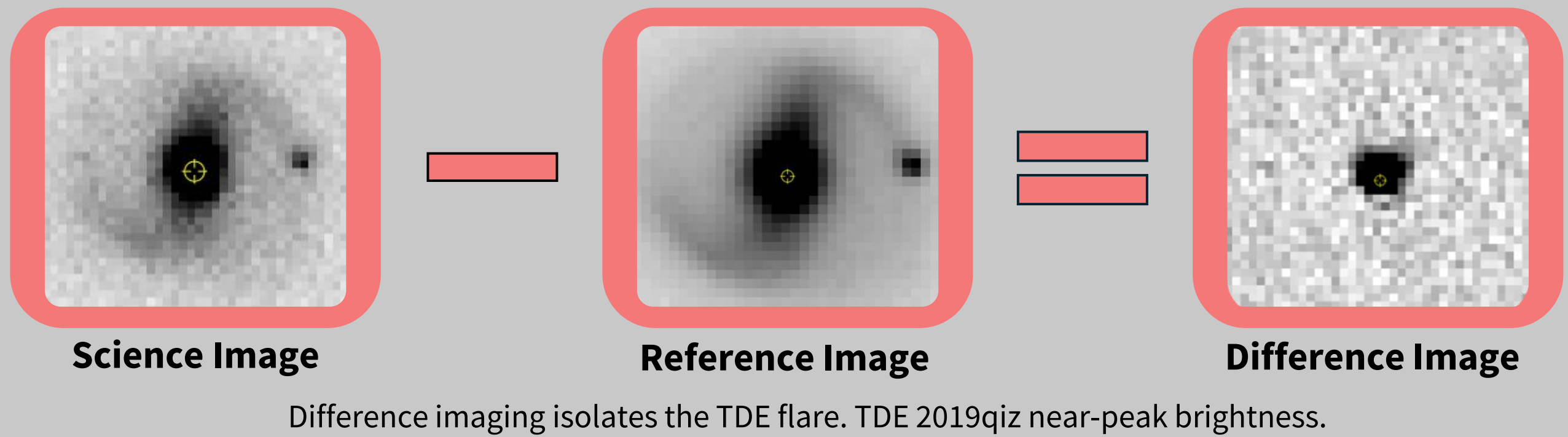
Results

Our pipeline correctly recovers both known offset TDEs currently in our sample. No new candidates have been identified yet, with only around 20% of known TDEs analyzed so far.

14 TDEs, 30 AGN

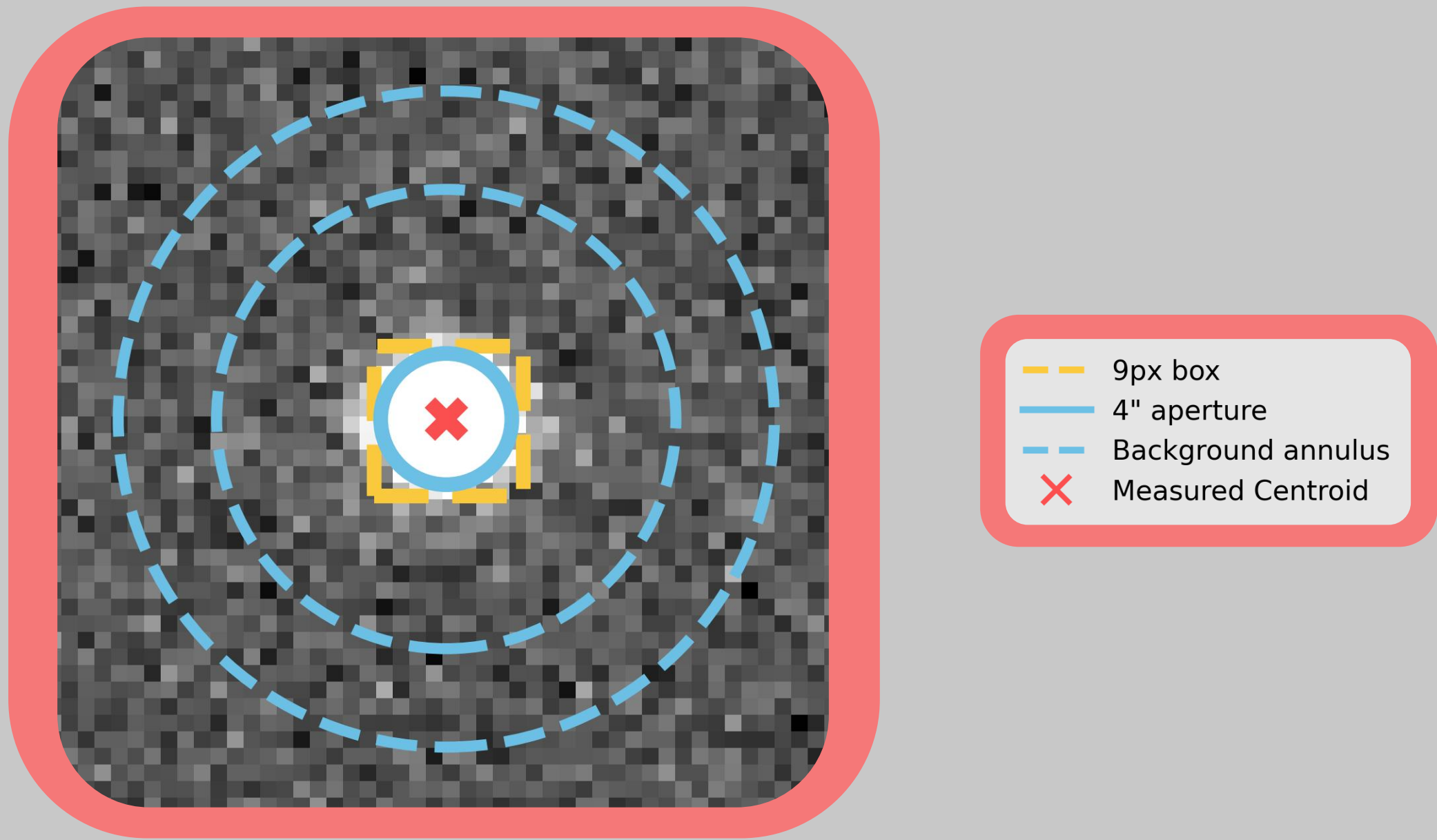
Methods

- Measuring host-galaxy offsets across known TDEs to reveal wandering black holes we can't otherwise detect
- ZTF (Zwicky Transient Facility), a wide-field optical survey of the northern sky: nearby, optically discovered TDEs & AGN comparison [7]
- Difference imaging **isolates the TDE flare** from the host galaxy



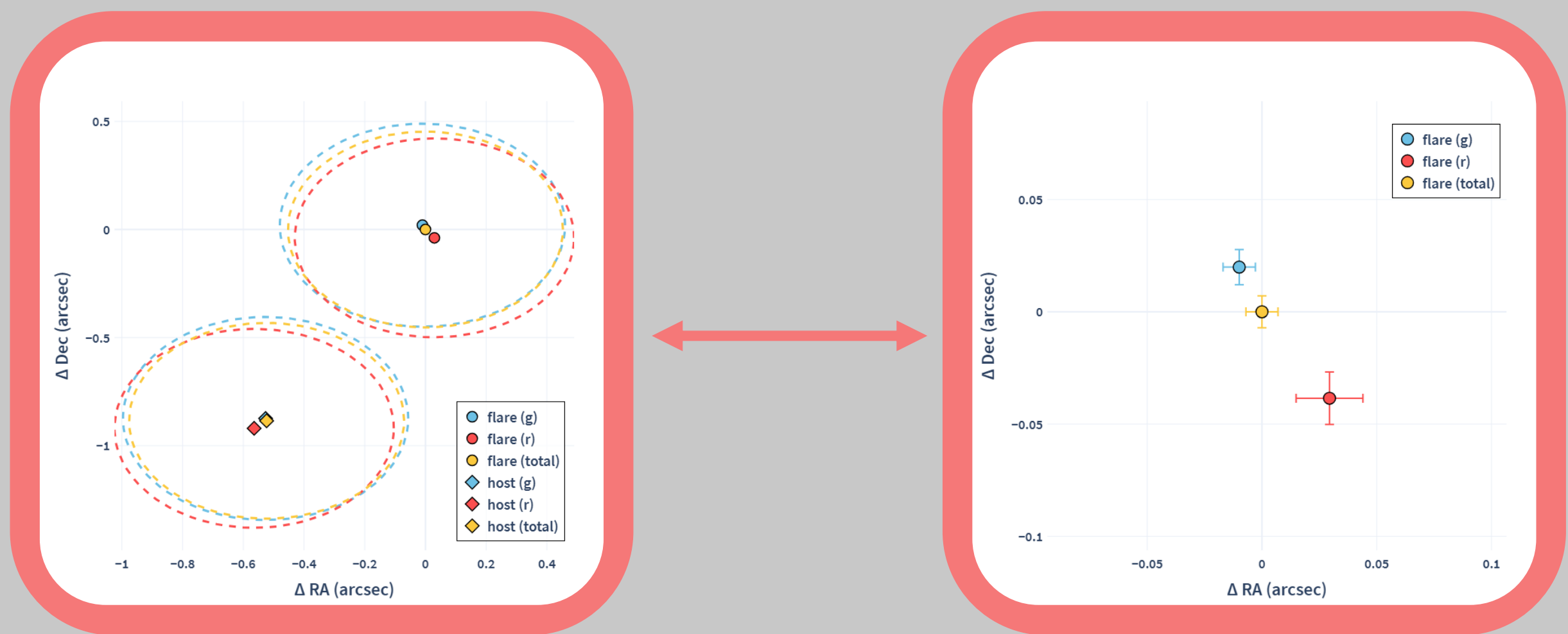
Difference imaging isolates the TDE flare. TDE 2019qiz near-peak brightness.

Example Difference Image:



TDE 2019azh, near-peak epoch

Systematic Scatter Outweighs Statistical Uncertainty:

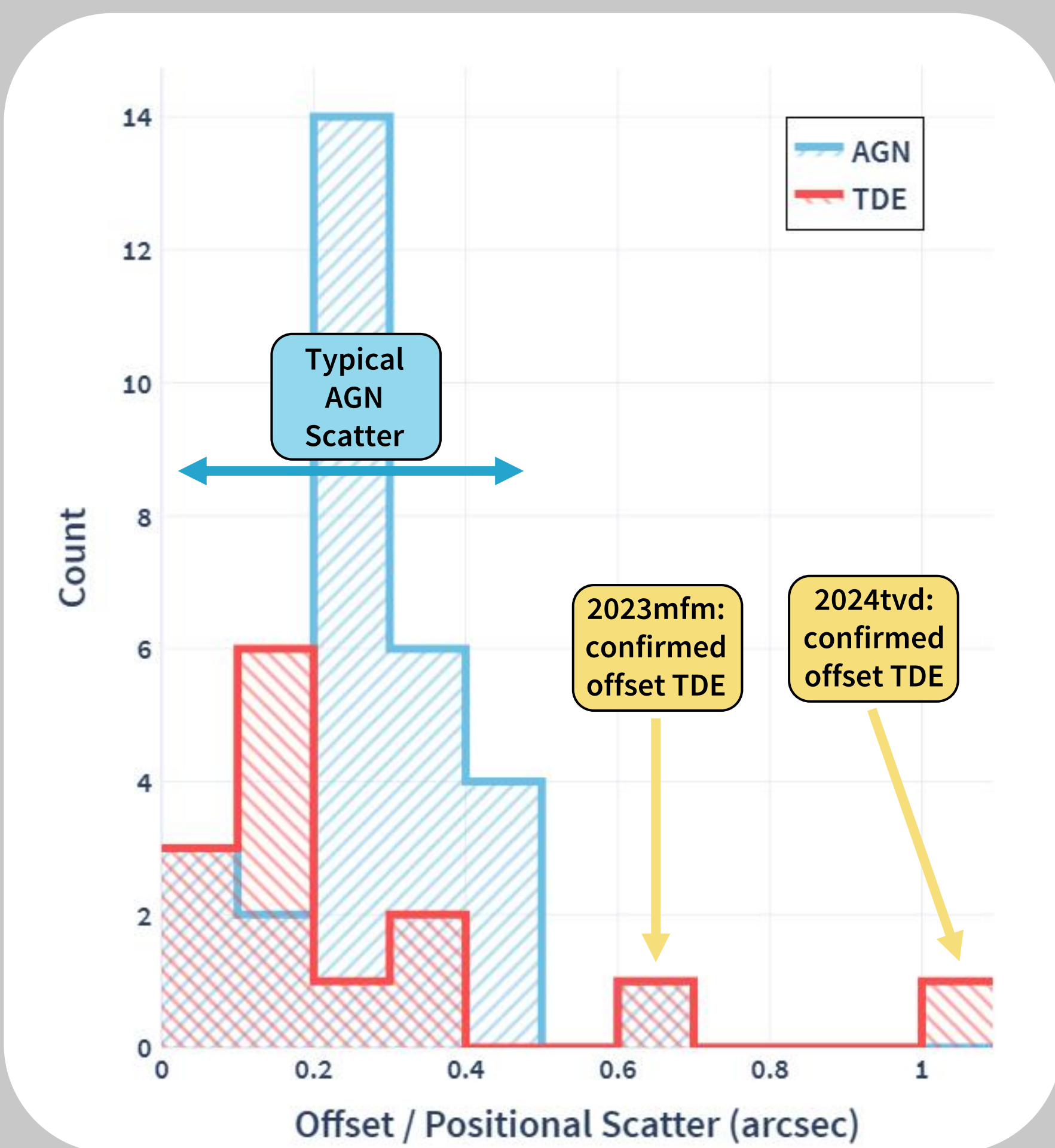


2024tvd, a confirmed off-nuclear TDE. Dashed circles show systematic uncertainty (AGN scatter) per filter.

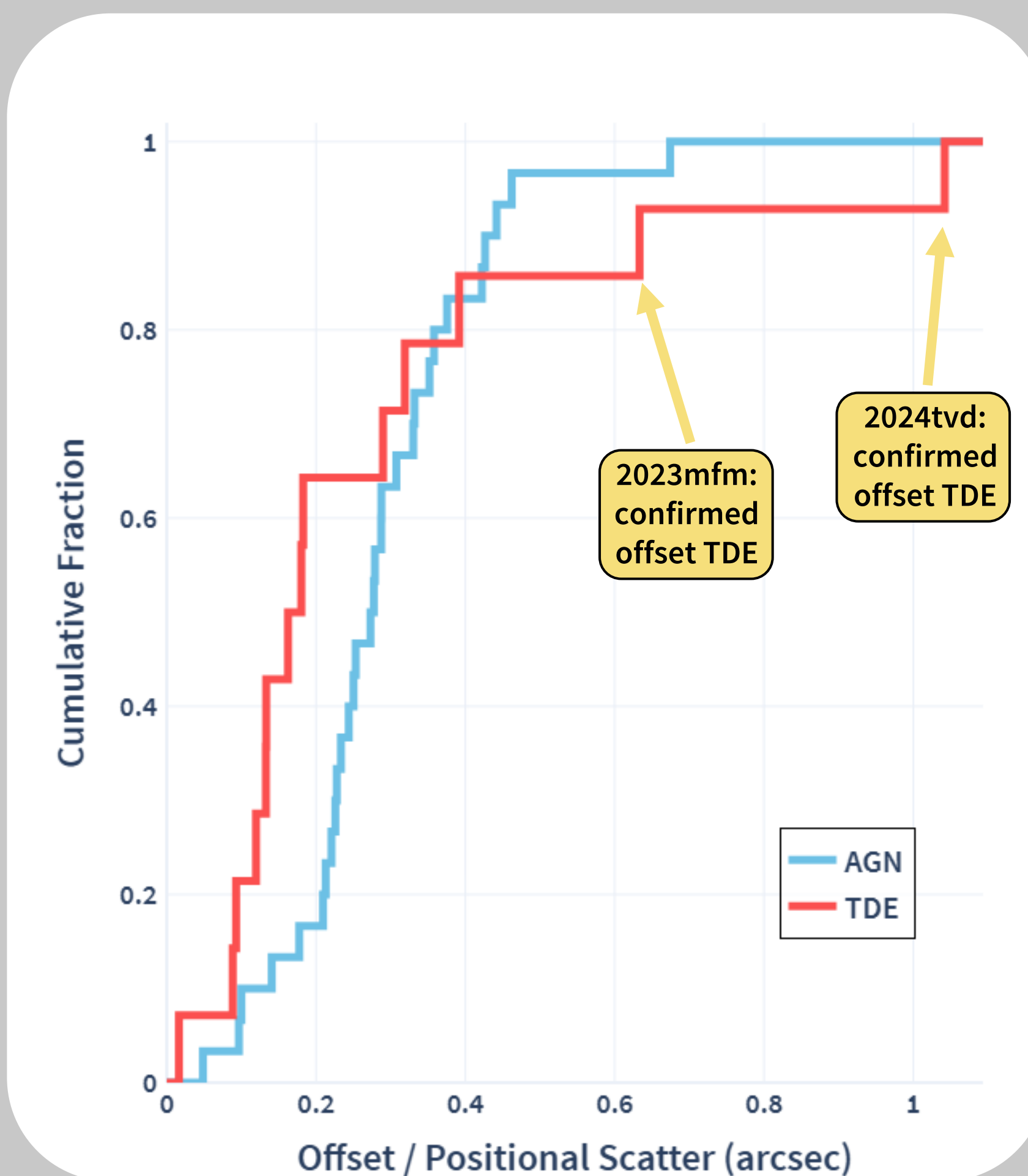
Zoomed: statistical uncertainty (SEM on mean position) is far smaller than systematic uncertainty

- 2D Gaussian centroiding** measures the flare's position in each image
- S/N vs. aperture radius slope cutoff removes images with poor centroid fits**
- Box size (9px) and aperture radius (4") to minimize positional scatter**
- Surviving positions combined across many images into one weighted mean position**
- AGN used as a control sample** to establish measurement noise floor

Beyond the Noise: Where TDEs Actually Land



TDE offsets (red) vs. AGN measurement scatter (blue), distinguished by color and hatch direction



Cumulative fraction of AGN scatter and TDE offsets below each value

95% of AGN positional scatter falls below 0.5". TDE offsets above this threshold are candidate offset TDEs, needing further follow-up

Future Work

- Complete analysis of full known optical TDE sample (~100) [3]
- Refine host galaxy position measurement by testing additional methods

Precise offset measurement should complement existing TDE classification, since a host-galaxy offset, a key signature of wandering black holes, has not previously been measured this carefully.

Acknowledgments

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References

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Top Right Figure (artist's concept of a TDE): Image credit: DESY, Science Communications Lab. Zwicky Transient Facility (ZTF)
Middle Figure (Hubble image of AT2024tvd): Image Credit: NASA, ESA, STScI, Yuhang Yao (UC Berkeley); Image processing: Joseph DePasquale (STScI)